## Detailed description of the design including the CONTROL treatment and the POLICY treatment. Please particularly outline the intervention in the POLICY treatment in detail.

The main research question is: "What is the impact of behavioral interventions on real-world support for carbon pricing?"

Our treatment involves providing more detailed information regarding the allocation of carbon tax revenues; specifically, the name and details of the institution that would manage the revenues. A carbon tax generates public revenues, and the allocation of these funds by policy-makers influences public support for the tax (Siegerink et al., 2024; Dominioni and Heine, 2019). Studies indicate that support tends to increase (Carattini et al., 2017) when society perceives the tax as environmentally effective (see, e.g., Klenert et al., 2018; Kaufman, 2019), especially when the revenues are used to finance green initiatives (Dominioni and Heine, 2019). We hypothesize that increasing transparency on revenue allocation will enhance the perceived environmental effectiveness of the tax.

In the CONTROL treatment, participants will be told that revenues will be:

> "*directed to research and development for low carbon alternatives for the sector*".

In the POLICY treatment, participants receive more information on the way revenue allocation will work. These participants will learn that revenues will support the Innovation Center for the U.S. Dairy, a private organization created by farmers that includes 28,000 farm families and dairy importers. The organization's mission is to develop sustainable solutions, focusing on reducing GHG emissions (including methane) and adopting renewable energy. Information on the effectiveness of such innovations in mitigating GHG emissions will also be provided. In particular, participants will be told that revenues will be:

> *"directed to research and development for low carbon alternatives for the sector. In particular, the revenue would be allocated to the Innovation Center for U.S. Dairy. The Center is a private organization founded by farmers, involving 28,000 dairy farm families and importers, which aims to offer advance solutions to build a more sustainable sector. The donation will go towards achieving greenhouse neutrality by 2050 (US Dairy Net Zero Initiative). The developing and integration of technologies into agricultural and livestock practices are critical for improving sustainability. These investments are very cost-effective in reducing greenhouse gas emissions (compared to other potential uses of the carbon price revenues). You can learn more about the U.S. Dairy Net Zero Initiative here"*.

To assess the impact of behavioral interventions on public support for carbon pricing, 350 participants will be recruited through Prolific (an online platform) and randomly assigned to either a control group or a treatment group receiving a behavioural intervention. The experiment uses a between-subjects design.

Both groups will be exposed to an introductory section where information about carbon taxes and GHG emissions, as well as compensation details for participating in the study will be provided. Each participant receives a flat payment of USD 5.5; depending on their decisions participants may retain the full amount or

only the base remuneration for a survey of this length under PROLIFIC`s terms (USD 2). Finally, a "cheap talk" script will clarify that there are no right or wrong answers, helping to prevent desirability bias.

Participants will have to make two decisions:

1. **Revealed support for carbon pricing outside the experiment**: participants will have the option to support the introduction of a carbon price by signing a petition on iPetitions.com, addressed to the Administrator of the Environmental Protection Agency and the Secretary of the US Department of Agriculture. The petition advocates for implementing a carbon tax with revenues directed to a private research and innovation centre[1]. In the POLICY intervention, additional details about the research and innovation centre will be provided (see below). To facilitate accurate tracking, participants will be asked to include their Prolific ID when signing even if this information will be only available to the research team. Upon completion of the experiment, the petition (with participant names) will be sent to the stated institutions to avoid deception.

2. **Revealed support for carbon pricing within the experiment**: following the first decision, participants will be asked whether they support the introduction of a carbon tax[2]. If they provide an affirmative response to this second decision, they will forgo part of their participation payment to stablish a carbon price. The amount reflects the impact of a carbon tax on milk expenditure, based on:

- average annual U.S. per capita fluid milk consumption of 15.12 gallons (USDA, 2023);
- a CO2 equivalent footprint of 3.882 kg CO2e per gallon of milk (Mazzeto et al., 2022); and
- a reference carbon price that incorporates all social costs of $120 per tonne of CO2 (EPA, 2023).

This yields an estimated increase in milk expenditure of 7 USD per year due to the carbon pricing scheme. Because participants cannot receive a pay-off below USD 2.0 and the participation fee is set at USD 5.5, the maximum impact on earnings due to support for carbon taxation is USD 3.5. This reflects the impact of the carbon price on their milk bill over six months.

Revenues from this choice (USD 3.5 x number of supporters) in both the CONTROL and POLICY treatments will be donated to the Innovation Center for U.S. Dairy.

The first decision will be used for comparisons across designs. Since question order is not randomized, i.e. participants will always answer decision #1 first, we expect no cross-contamination from decision #2.

After the experiment, if time permits, participants will complete a brief questionnaire on dairy consumption habits (1 question), attitudes toward the agricultural sector (4-5 questions), and basic demographic information (depending on Prolific's data availability).

**REFERENCES**

– Carattini, S., Baranzini, A., & Lalive, R. 2018. Is taxing waste a waste of time? Evidence from a supreme court decision. *Ecological Economics*, 148, 131–151.

[1] The text of the petition, which is the same for the CONTROL and POLICY groups is included as Annex 1.
[2] The wording of the question will be modified depending on their response to the first decision.

- Dominioni, G., Heine, D., 2019. Behavioural economics and public support for carbon pricing: A revenue recycling scheme to address the political economy of carbon taxation. *Eur. j. risk regul.* 10, 554–570. https://doi.org/10.1017/err.2019.44
- EPA (Environment Protection Agency), 2024. *Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances*. Retrieved from https://www.epa.gov/system/files/documents/2023-12/epa_scghg_2023_report_final.pdf
- Kaufman, N., 2019. *What you need to know about a federal carbon tax in the United States*, Finance & Economics. Columbia Center on Global Energy Policy, New York.
- Klenert, D., Mattauch, L., Combet, E., Edenhofer, O., Hepburn, C., Rafaty, R., Stern, N., 2018. Making carbon pricing work for citizens. *Nature Clim Change* 8, 669–677. https://doi.org/10.1038/s41558-018-0201-2
- Mazzeto, A.M., Falconer, S., Ledgard, S. 2022. Mapping the carbon footprint of milk production from cattle: A systematic review. *J. Dairy Sci.* 105, 9713–9725. https://doi.org/10.3168/jds.2022-22117
- Siegerink, V.E., Delnoij, J., Alpizar, F., 2024. Public preferences for meat tax attributes in The Netherlands: A discrete choice experiment. *Food Policy* 128, 102675. https://doi.org/10.1016/j.foodpol.2024.102675
- USDA., 2023. *Dairy products: Per capita consumption*, United States. Economic Research Service, U.S. Department of Agriculture.


## Detailed description of the outcome variable "real-world support for a price of carbon" and especially its real-world impact.

In this study, the first decision measures real-world behaviour beyond the experimental setting through a dichotomous (dummy) variable, indicating whether participants sign the iPetitions.com petition in support of the carbon tax. The real-world impact of this variable is demonstrated by submitting the completed petition signed by participating individuals to the designated recipient upon closure.

## Please indicate in 3-5 sentences whether and why the conditions for the shortcut IRB apply to your experiment.

Additional details on the payment levels:

Our experiment has two potential payment levels, depending on participant responses to the second decision:

- If they support the carbon price, they keep USD 2.0 (USD 5.5 minus USD 3.5).
- If they do not support the carbon price, they keep USD 5.5.

The USD 3.5 will be transferred to the Innovation Center for U.S. Dairy to support its NET ZERO initiative.

Assuming 50% support in the control group and 65% in the treatment group, the total PROLIFIC reward costs are estimated at USD 1220.63, with an additional USD 704.37 designated as a donation to the Innovation Center for US Dairy.

The following table provides detailed projections based on a total sample size of 350, evenly split across treatments:

| Group | Accept payment | Expected share | Compensation for self | Total payments to participants | Total payments to Innovation Dairy Center |
|---|---|---|---|---|---|
| Control | YES | 0.5 | 2.0 | 175.00 | 306.25 |
| | NO | 0.5 | 5.5 | 481.25 | 0.00 |
| Policy | YES | 0.65 | 2.0 | 227.50 | 398.12 |
| | NO | 0.35 | 5.5 | 336.88 | 0.00 |
| TOTAL | | | | **1 220.63** | **704.37** |

1. To avoid any form of deception, the total amount collected from supporters will be donated to the Innovation Center for U.S. Dairy.
2. Only participants´ Prolific IDs will be recorded, ensuring anonymity within the experiment. Personal information (name) will only be registered within the iPetitions.com platform.
3. Beyond group assignment (control or treated), no further random procedures are involved.
4. The experiment design includes no measures of physical or psychological state and poses no related risks.
5. Clear instructions on experiment duration and participant guidelines are provided, with no repetitions, interactions, or randomization procedures.
6. Incentive compatible payments are designed to cover participants' opportunity costs and adhere to PROLIFIC's minimum payment standard (USD 1.88, with a practical minimum of USD 2). Negative earnings are not possible.

## Please specify your incentive structure.

The experiment employs a well-defined incentive system. All participants receive USD 5.5 as a flat payment for participation. In Stage 2, participants are given the option to contribute USD 3.5, equivalent to the impact of a six-month carbon tax on their milk expenditure. Depending of their decisions, they will retain either the total endowment of USD 5.5 or just USD 2.0.

To prevent any form of deception, the total amount collected from the behavioral intervention group will be donated to the Innovation Center for U.S. Dairy.

## Is the treatment effect identifiable using OLS, a probit/logit, or an ordered-logit model?

Yes.

## What is the estimated length of the experiment? (excl. policy survey)

8.0 minutes.

## Is your proposal using incentives?

Yes.

## What is the average payment per participant (including flat fee + potential bonus payment)?

USD 3.49 (with an additional expected payment of USD 704.37 to the Innovation Center of the US Dairy).

## Which experimental software do you plan to use?

Qualtrics.

## What is your expected effect size? (Cohen's d for continuous outcome variables and Cohen's h for shares/proportions)'

0.2

**ANNEX I. Petition text to be used**

Mr. Michael S. Regan
Administrator
Environmental Protection Agency
1200 Pennsylvania Avenue, N.W.
Washington, DC 20460

Regan.Michael@epa.gov

Mr. Tom Vislack
Secratary of Agriculture
U.S. Department of Agriculture
1400 Independence Ave., S.W.
Washington, DC 20250

agsec@usda.gov

**Subject:** Support for introducing a carbon price in the U.S. diary sector with revenues used to to support decarbonisation

The undersigned urge policymakers to introduce a carbon price in the U.S. dairy sector with the aim of reducing its environmental impact through decarbonisation.

Revenue generated from this carbon price should be allocated to fund research and development for low carbon alternatives for the sector.

A carbon price would allow consumers to make purchase decisions that reflect the environmental cost of production, encouraging shifts towards more sustainable consumption patterns. It will also incentivize producers to adopt technologies and practices that reduce greenhouse emissions and improve their environmental footprint. This approach aligns with broader environmental goals and empowers consumers to play an active role in decarbonisation and sustainability.

By signing this petition, we express our commitment to creating a more sustainable food system. We believe that a carbon price in the U.S. dairy sector would contribute to reducing its environmental footprint and encourage greener choices for both producers and consumers.